```
Zimbabwe, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.276             -31878.1            -31864.4                 0
     2 20263017             5000           0.276             -31877.2            -31866.1                 0
     3 20263018             5000           0.275             -31873.1            -31865.7                 0
     4 20263019             5000           0.275             -31875.1            -31866.8                 0
                                      file
 mcmc_zimbabwe_step2_chain1_2026-10-19.Rdata
 mcmc_zimbabwe_step2_chain2_2026-10-20.Rdata
 mcmc_zimbabwe_step2_chain3_2026-10-20.Rdata
 mcmc_zimbabwe_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
     age.25.to.49.hiv.mortality.multiplier.1      age.5.to.14.hiv.mortality.multiplier.0
                                      1.811                                      1.748
             hiv.specific.mortality.rates.2                  female.to.male.multiplier
                                      1.691                                      1.615
      age.20.to.29.transmission.multiplier.0      age.30.to.39.transmission.multiplier.0
                                      1.614                                      1.513
             hiv.specific.mortality.rates.1      age.0.to.14.hiv.mortality.multiplier.2
                                      1.507                                      1.479
      age.30.to.39.transmission.multiplier.3          over.50.hiv.mortality.multiplier.1
                                      1.444                                      1.424
                                    trate.3      age.20.to.29.transmission.multiplier.3
                                      1.402                                      1.381
      age.0.to.14.hiv.mortality.multiplier.1   age.50.and.over.transmission.multiplier.3
                                      1.376                                      1.365
     age.15.to.24.hiv.mortality.multiplier.1                                    trate.4
                                      1.357                                      1.333
                                    trate.2                                    trate.0
                                      1.333                                      1.310
age.65.to.79.mortality.slope.multiplier.male      age.40.to.49.transmission.multiplier.0
                                      1.289                                      1.283
      age.15.to.19.transmission.multiplier.3             male.hiv.mortality.multiplier.2
                                      1.225                                      1.203
age.45.to.64.mortality.slope.multiplier.male      age.15.to.19.transmission.multiplier.2
                                      1.189                                      1.156
                     log.OR.testing.intercept
                                      1.154

Merged simset:
 location simulations total.weight                                              file
 Zimbabwe        1000         1000 simset_zimbabwe_step2_chains1234_2026-10-20.Rdata
```

Zimbabwe step 2: likelihood

0
−10000
−20000
−30000

value

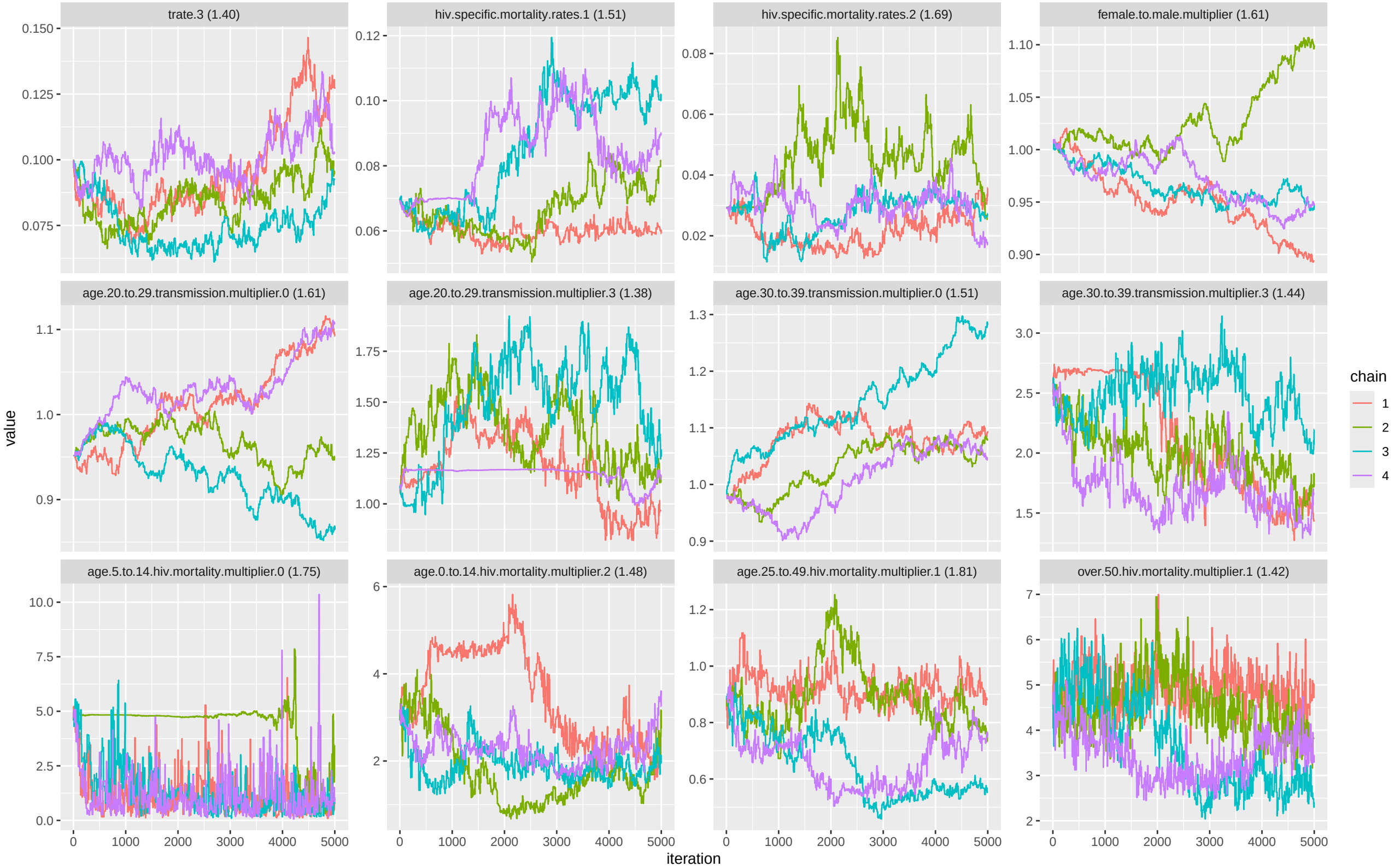
0
25000
50000
75000
100000

iteration

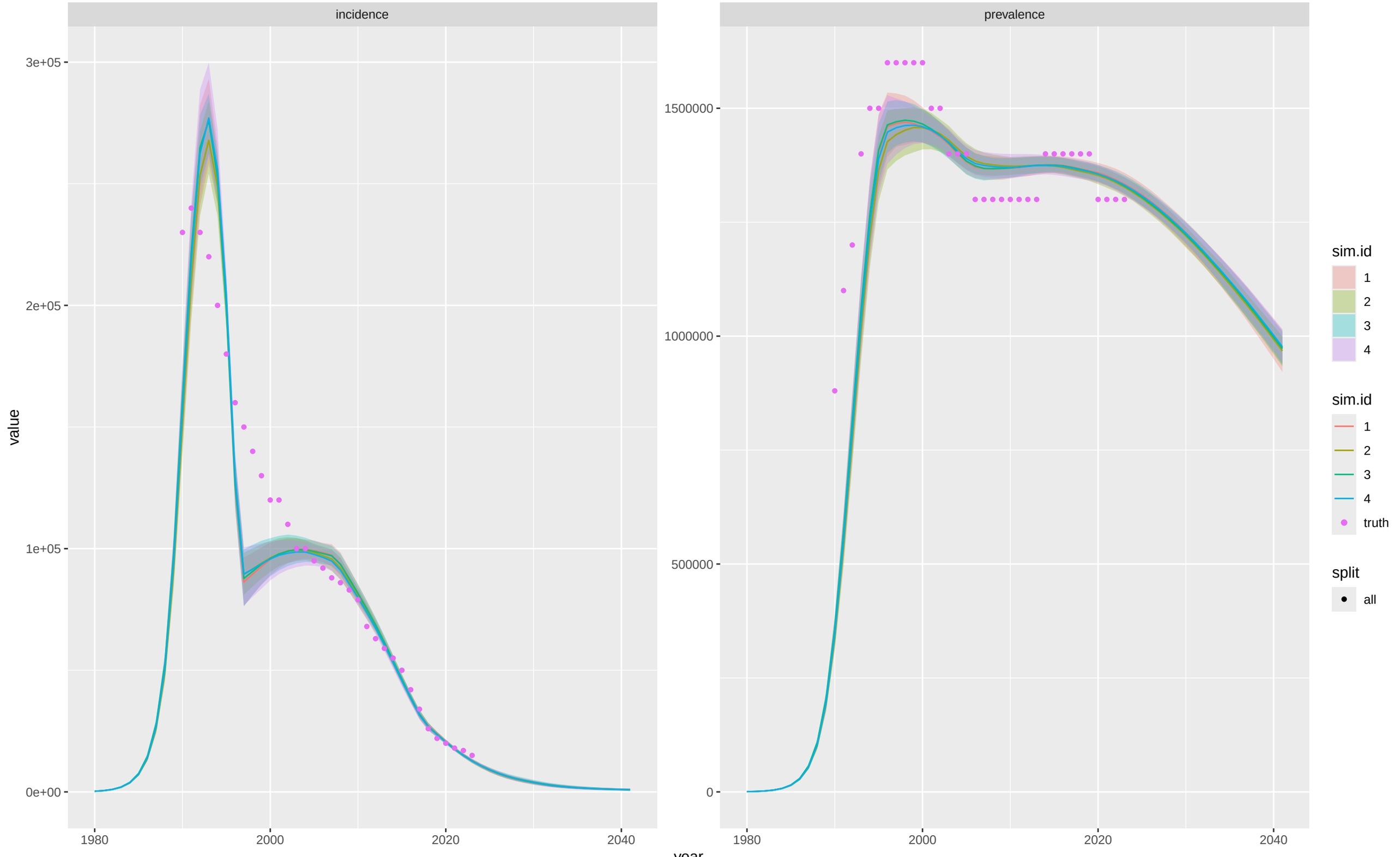
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

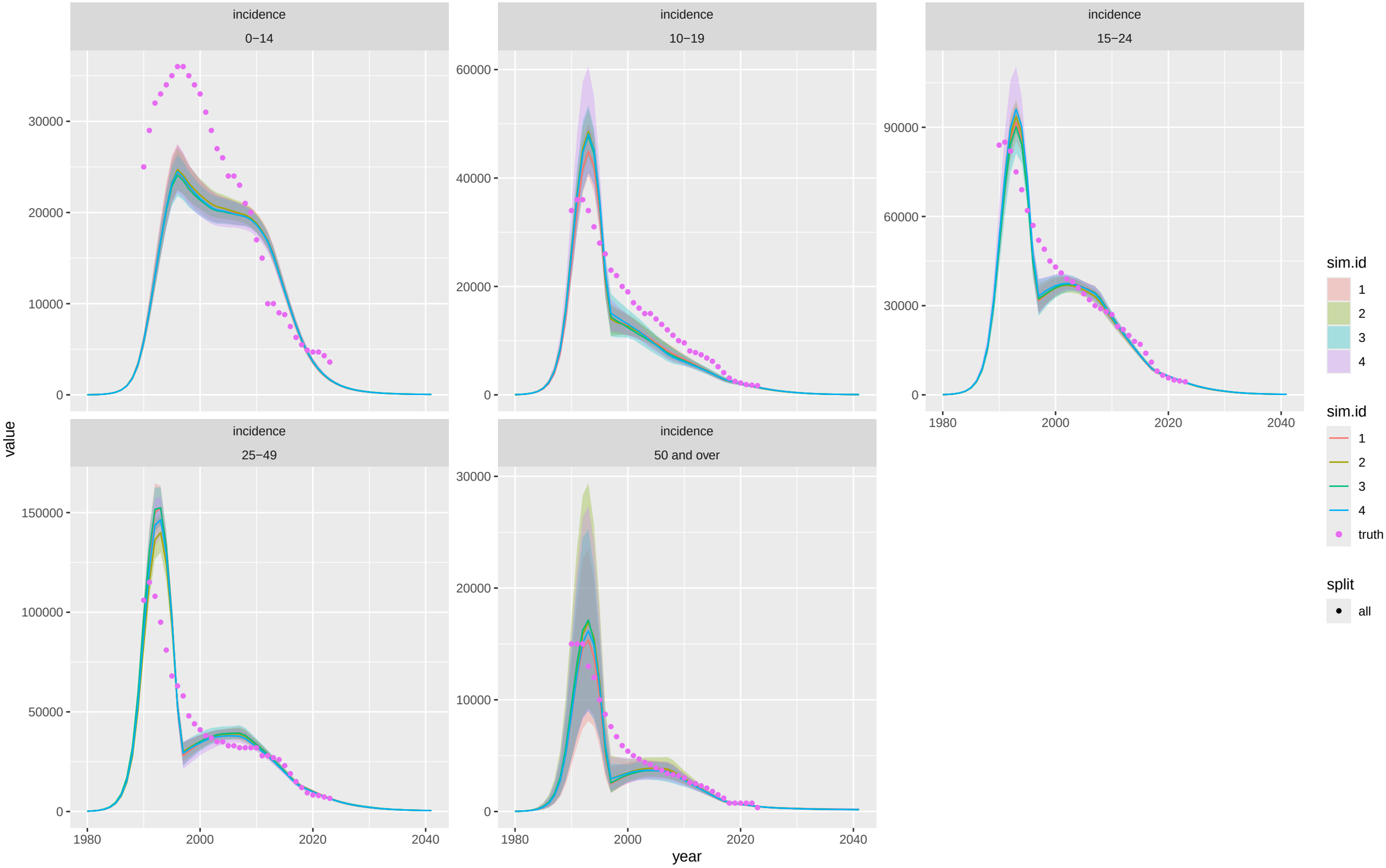
Zimbabwe step 2: trace



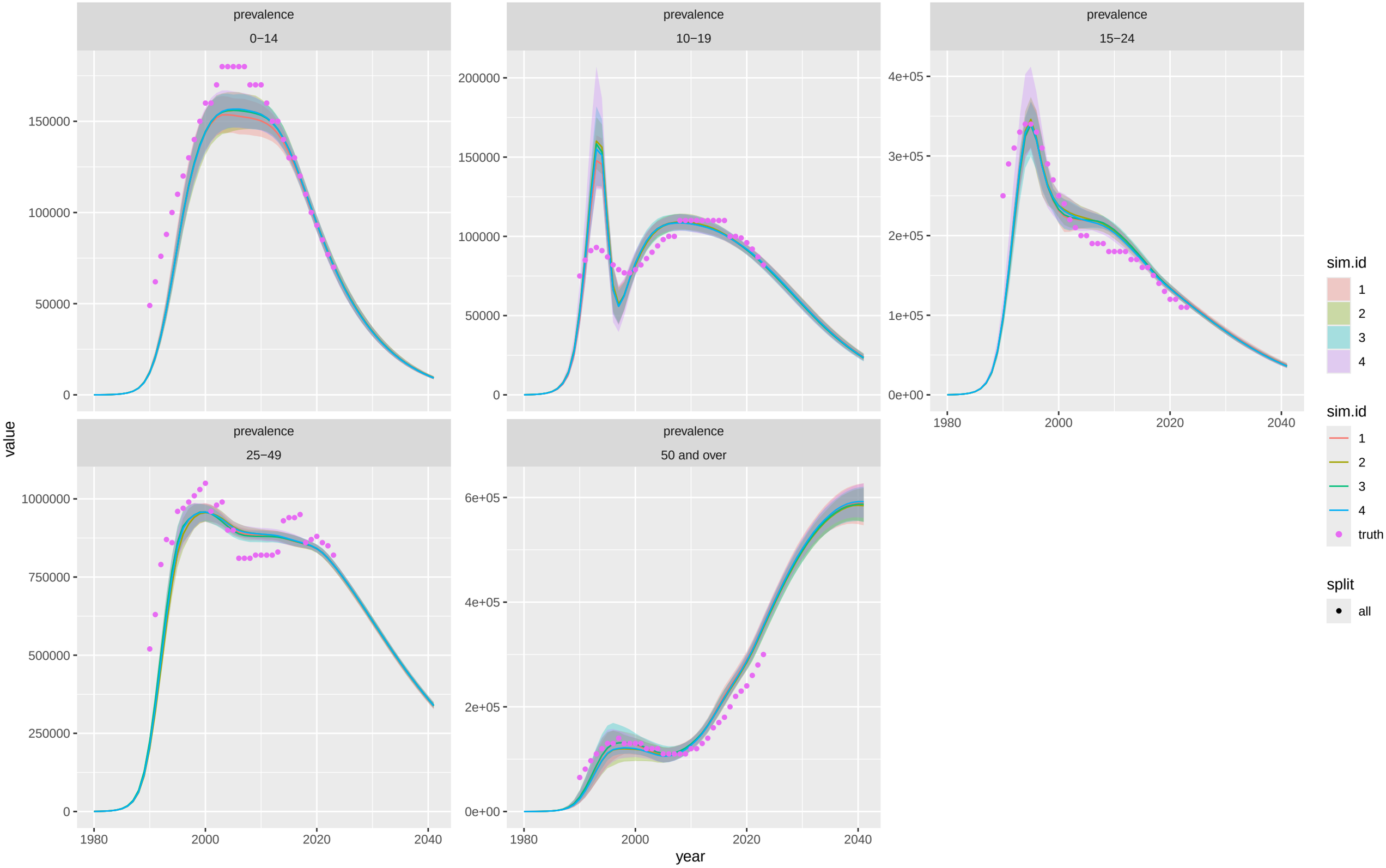
Zimbabwe step 2: fit (one color per chain)



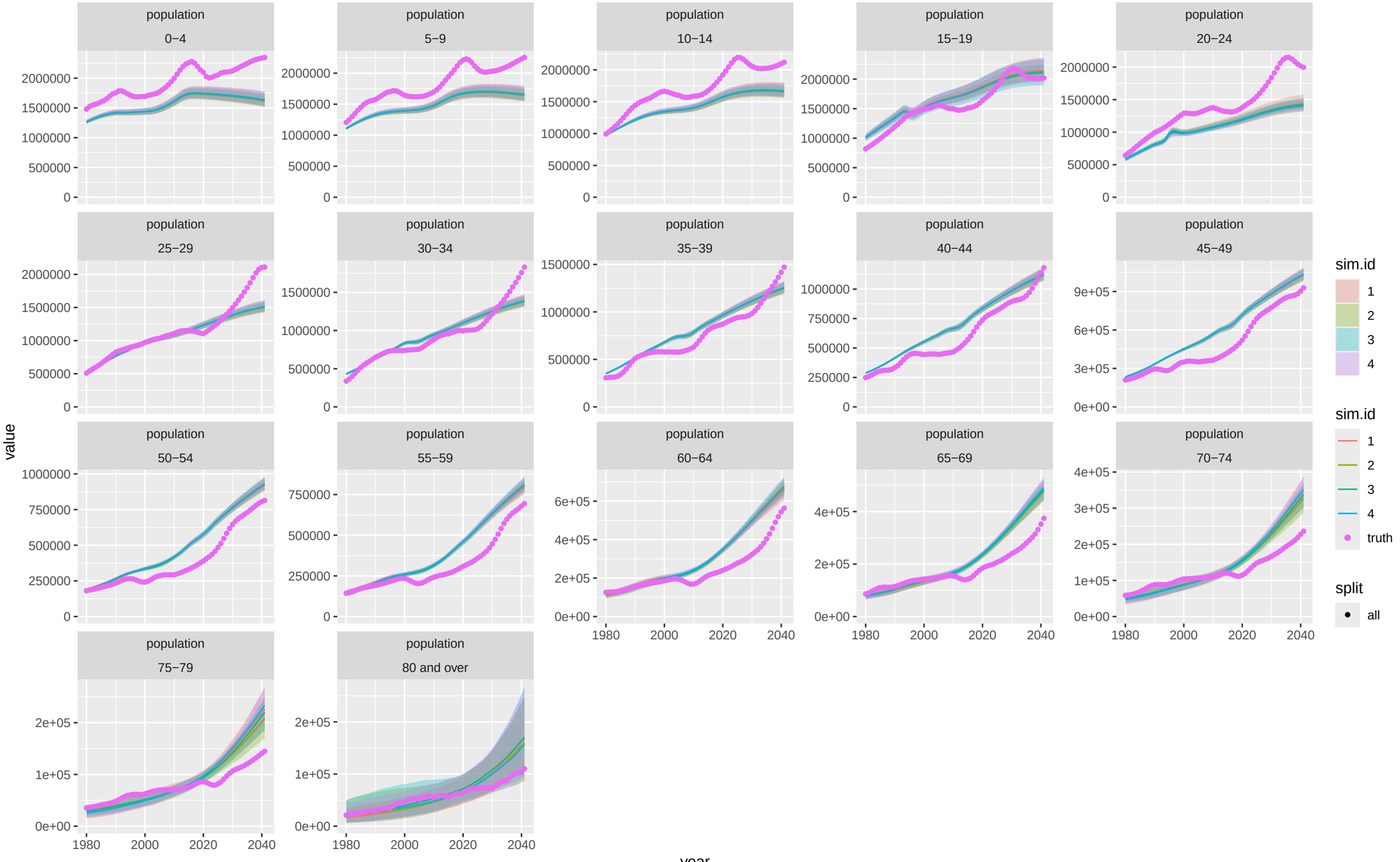
Zimbabwe step 2: incidence.by.age (one color per chain)



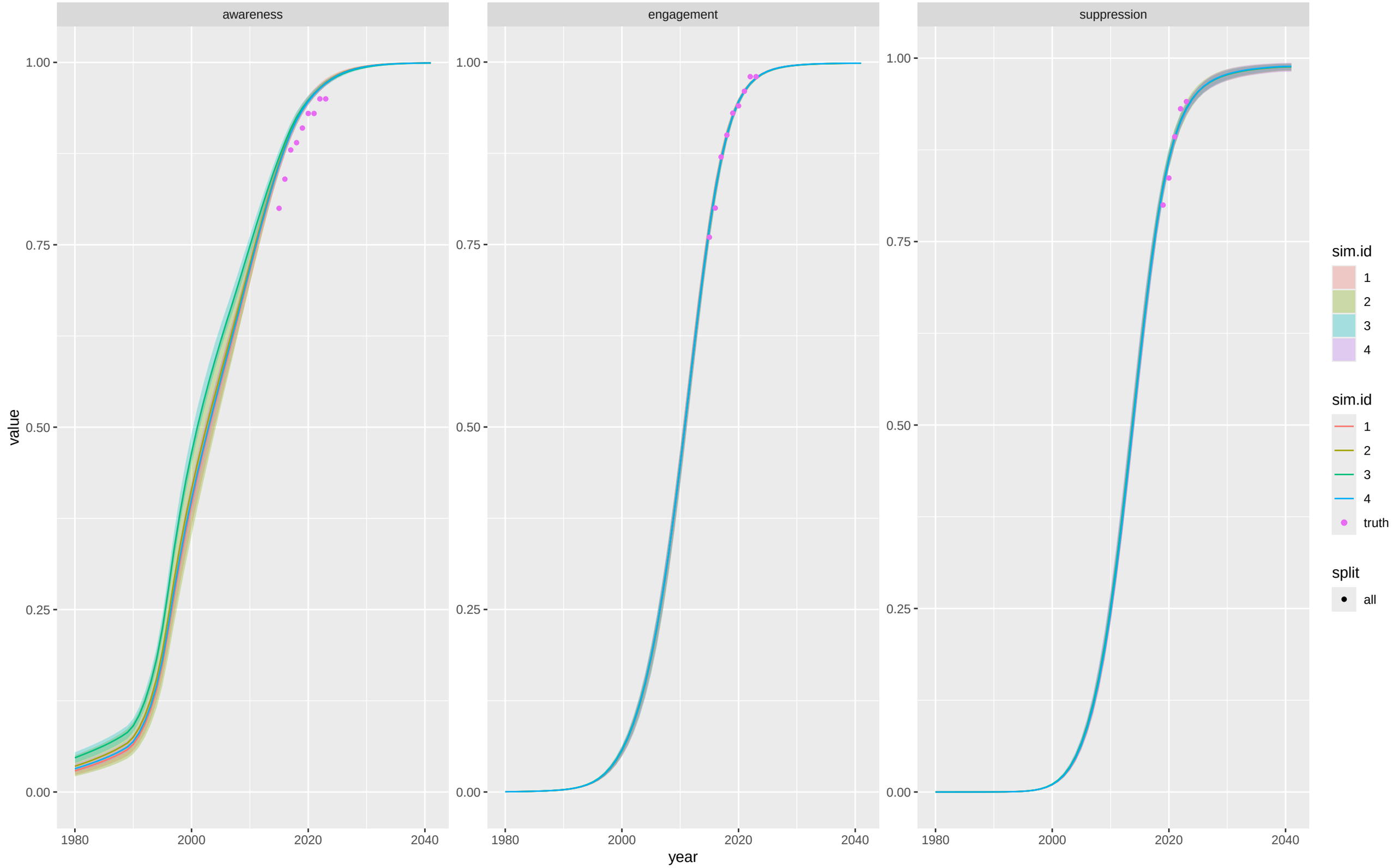
Zimbabwe step 2: prevalence.by.age (one color per chain)



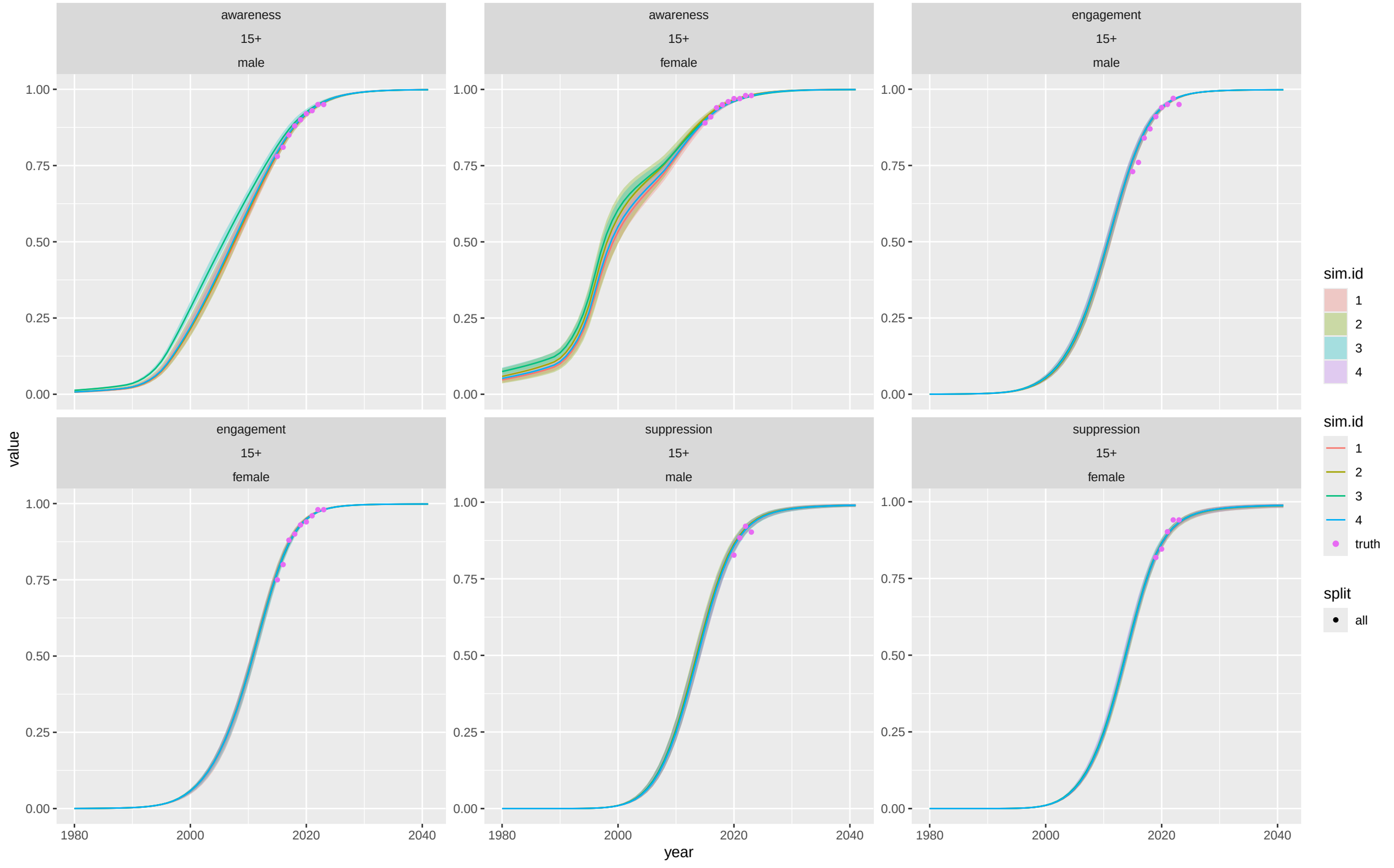
Zimbabwe step 2: population.by.age (one color per chain)



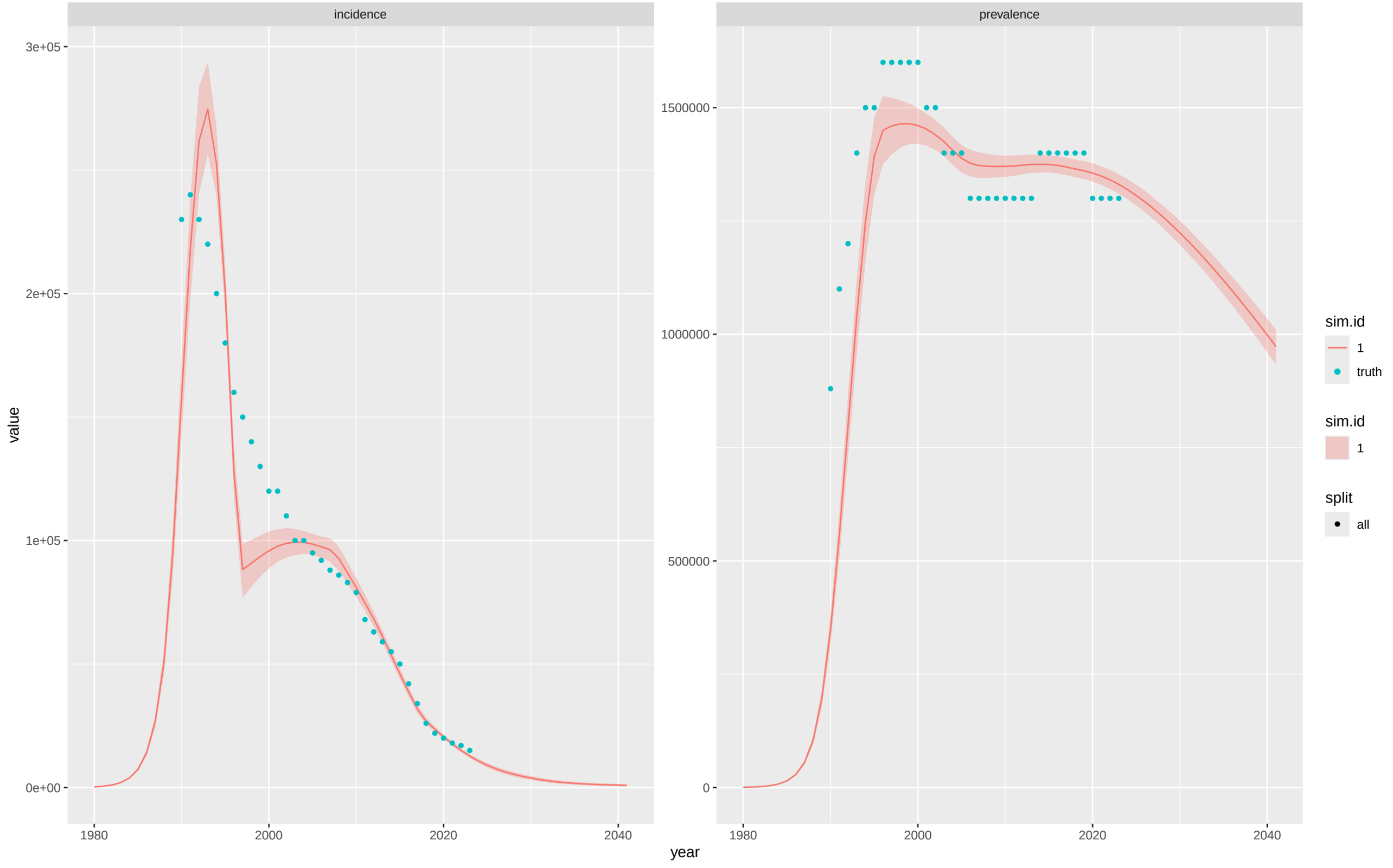
Zimbabwe step 2: cascade (one color per chain)



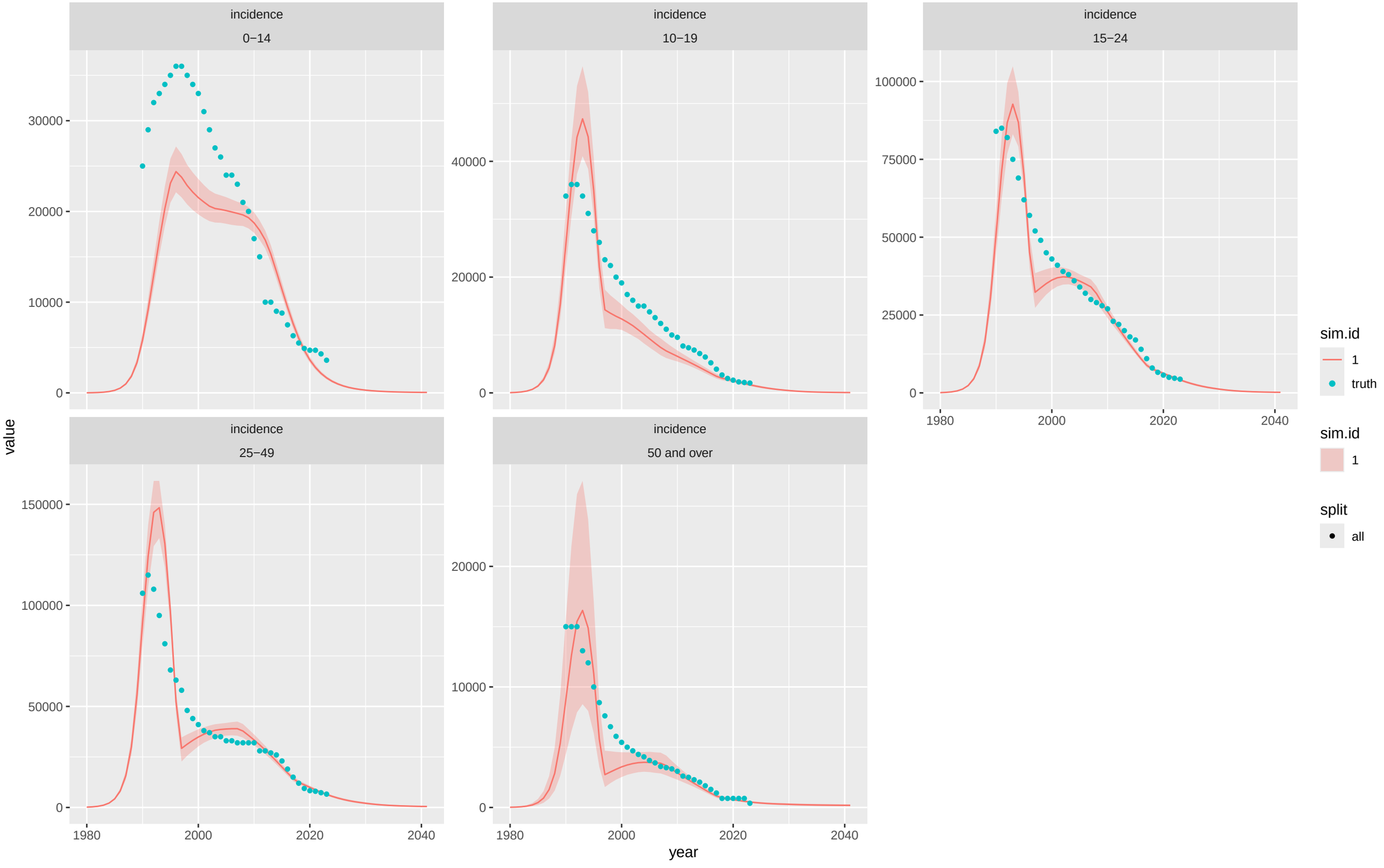
Zimbabwe step 2: cascade.by.age.sex (one color per chain)



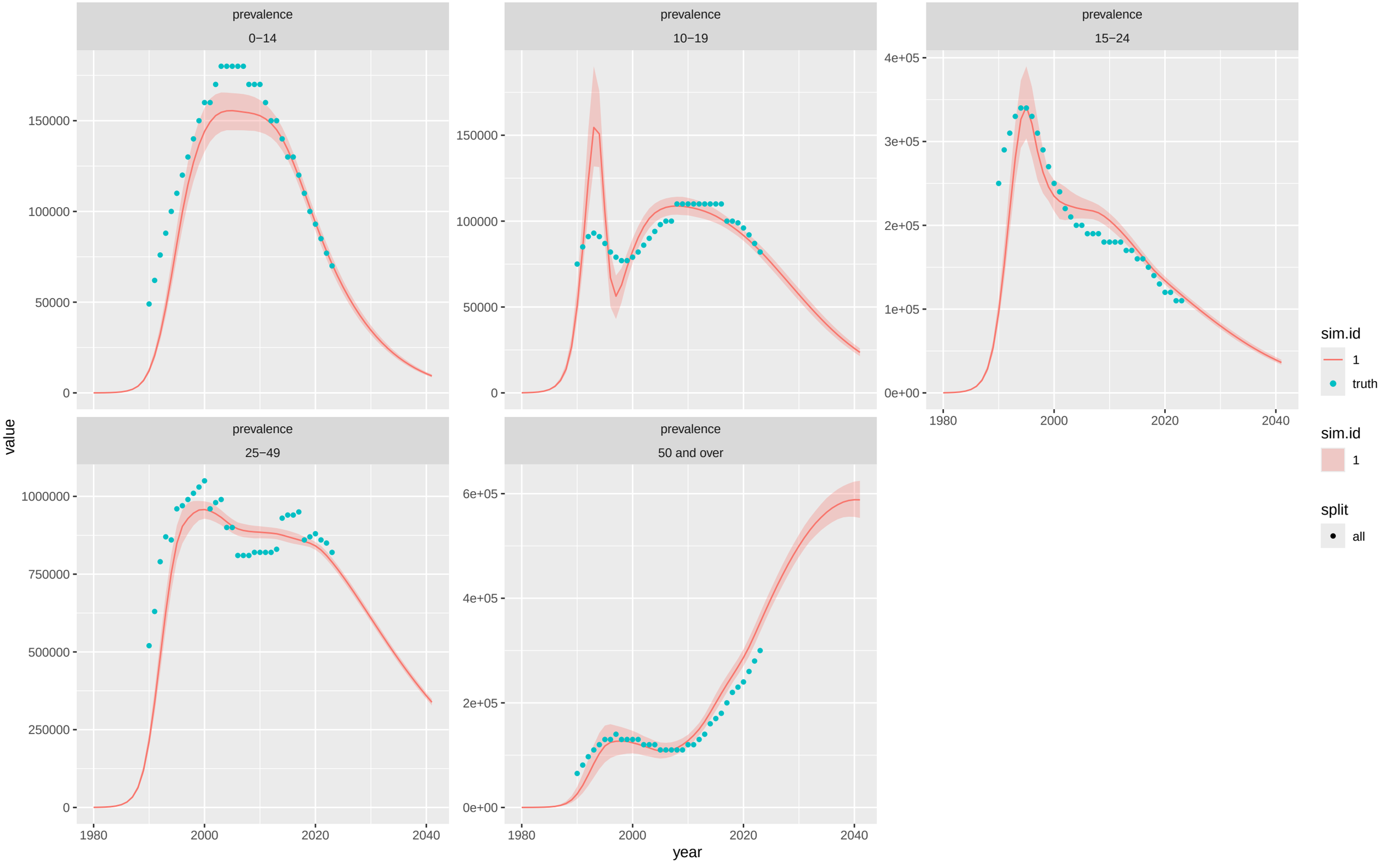
Zimbabwe: fit (merged Step 2 simset)



Zimbabwe: incidence.by.age (merged Step 2 simset)



Zimbabwe: prevalence.by.age (merged Step 2 simset)



Zimbabwe: population.by.age (merged Step 2 simset)

population 0−4
population 5−9
population 10−14
population 15−19
population 20−24
population 25−29
population 30−34
population 35−39
population 40−44
population 45−49
population 50−54
population 55−59
population 60−64
population 65−69
population 70−74
population 75−79
population 80 and over

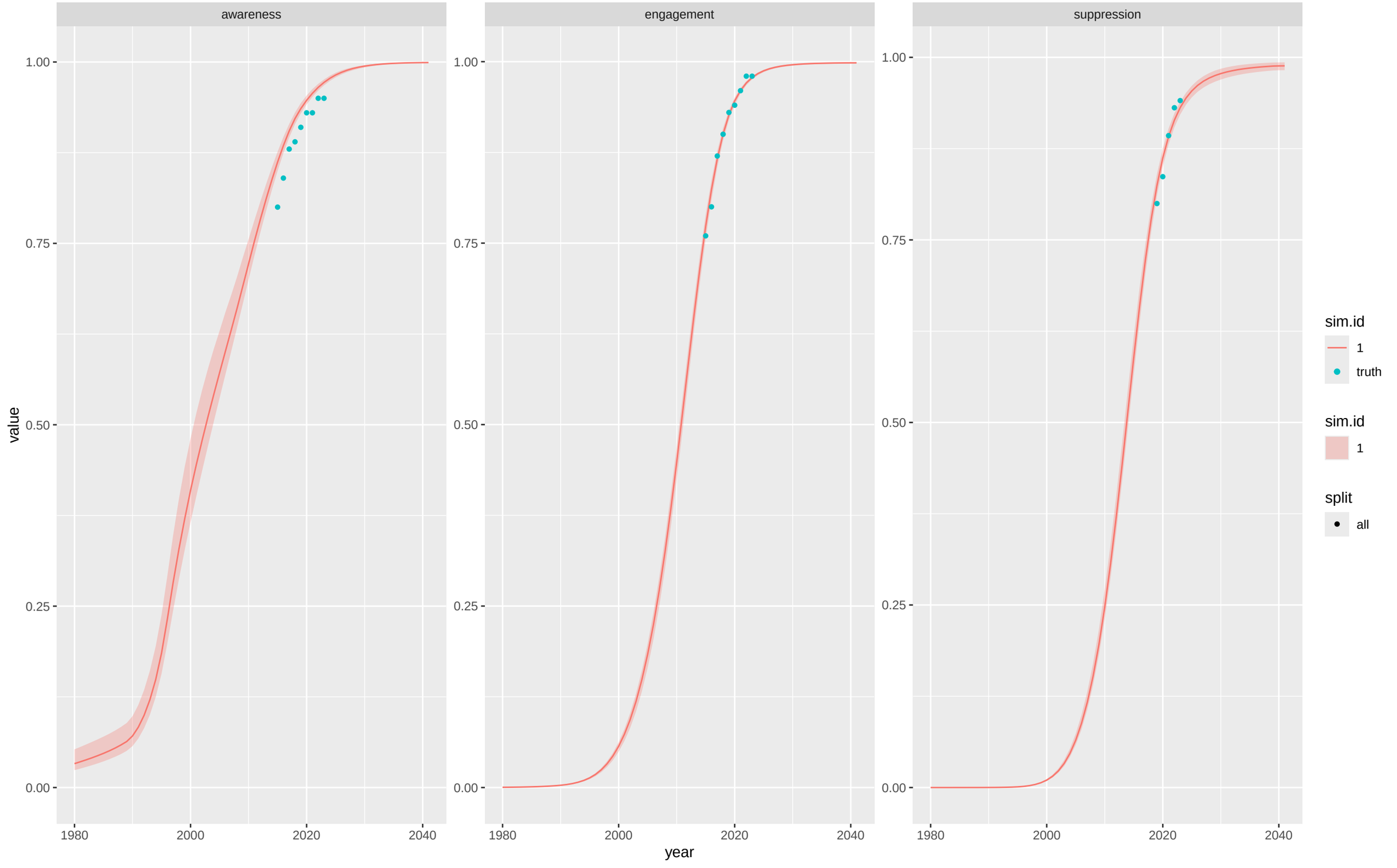
value

year

sim.id
1
truth

split
all

Zimbabwe: cascade (merged Step 2 simset)



Zimbabwe: cascade.by.age.sex (merged Step 2 simset)

